



UNIVERSITY OF PISA

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Global transport in soaring flights

Measurements, controls and open model questions

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Chapter 3 • 20-minute supervisor meeting

What motion should a stochastic model reproduce?

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f = |\mathbf{X}_f|.$$

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f}.$$

Windows stay within a retained segment. The main comparison covers $10 \leq \tau \leq 10^4$ s.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

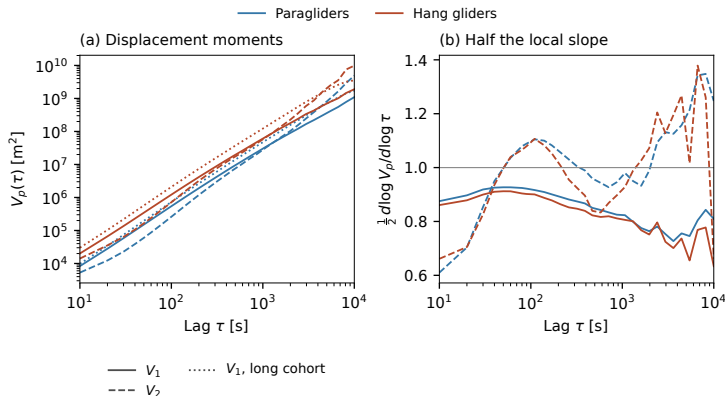
Two levels of evidence

Archive: 156,305 paragliders and 6,093 hang gliders.

Detailed diagnostics: 955 and 479 selected flights, one longest segment per flight, on a 10-s grid.

Archive baseline: equal segment weights. Subset V_1, V_2 : equal flight weights. Baseline quantiles and moments: pooled windows.

The MSD grows faster than linearly on the measured interval



	b_1	b_2
Para	1.702	2.028
Hang	1.662	1.985

$V_1 = \langle R^2 \rangle$; V_2 uses second differences.

The local slope bends. One fitted number does not establish a power law.

Subset; 10–10,000 s. Solid: V_1 ; dashed: V_2 ; dotted: V_1 , fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

Second differences cancel constant velocity; an intrinsic H remains open

$$A_2 \mathbf{r} = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad V_2 = \langle |A_2 \mathbf{r}|^2 \rangle.$$

Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2 \mathbf{r} = A_2 \mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

Conditional interpretation

If the centred residual has stationary increments and variance $K\tau^{2H}$,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

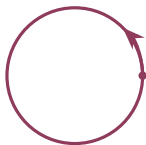
At $H = 1$ the prefactor vanishes. Our $b_2 \simeq 2$ cannot be read as a validated $H \simeq 1$.

Which residual and which scale interval could satisfy the assumptions needed to estimate H ?

At $\tau = 10^4$ s, V_2 retains only 107 para / 56 hang flights; it needs 2τ of continuous support.

Open and closed tasks change the growth of displacements

Open straight course



Closed circular course

Measured subset

	Task	N	b_1	b_2
Para	Open	391	1.752	1.968
	Closed	559	1.652	2.054
Hang	Open	171	1.702	1.893
	Closed	308	1.641	2.004

The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

Compare tasks at matched duration and region, using both physical lag and τ/T .

Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. [Exact circular example](#).

A single displacement scale predicts equal quantile slopes

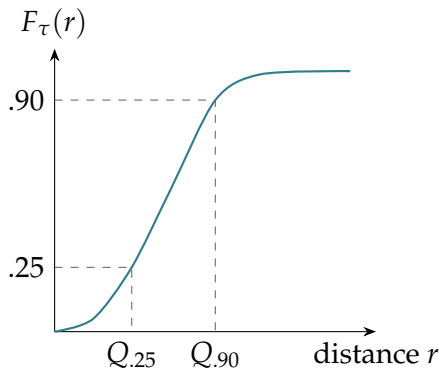
$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

An H -self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$

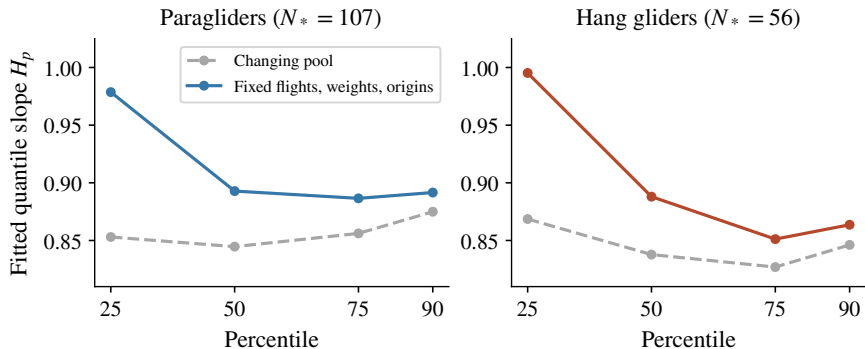
$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$H_p = H$ for every p ; $Q_{.90}/Q_{.25}$ stays fixed.



Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.

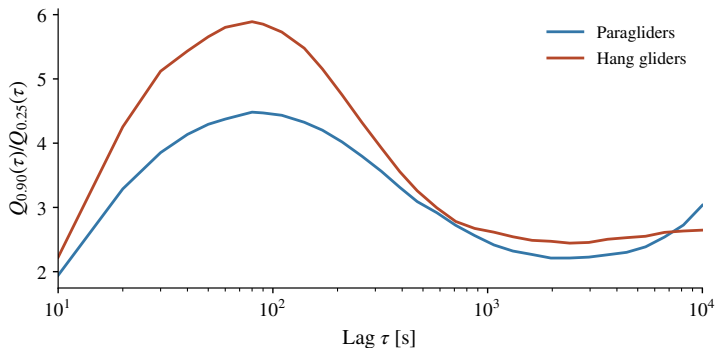
Controlling the population changes the quantile result



Fixed control: the same flights, equal flight weights and the same origins at every lag. The long cohorts contain **107 para / 56 hang flights** with segments $\geq 20,000$ s.

The changing para pool gives $H_{90} > H_{25}$; the fixed control gives the reverse. Intermediate weight/origin controls are in the [backup figure](#). Slopes are descriptive.

Relative spread changes, but the change is not monotone



$$\frac{d \log(Q_{.90}/Q_{.25})}{d \log \tau} = H_{.90}(\tau) - H_{.25}(\tau).$$

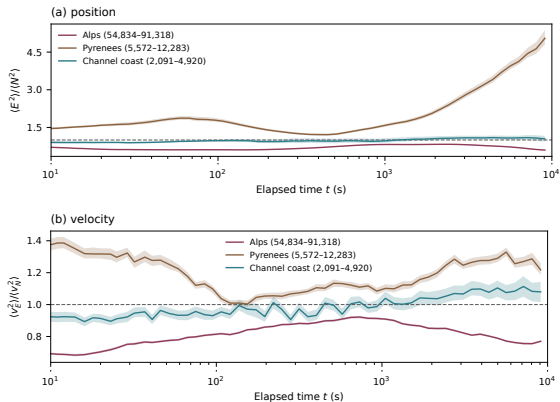
The ratio first rises, then falls, then rises again.

A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

How much of this shape evolution survives site–day uncertainty, route matching and phase conditioning?

Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

The Pyrenean archive has a strong east–west preference



$$A_u(t) = \frac{\langle u_E(t)^2 \rangle}{\langle u_N(t)^2 \rangle}.$$

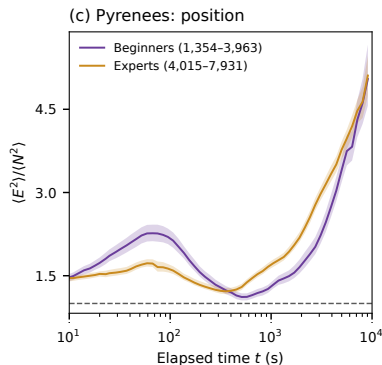
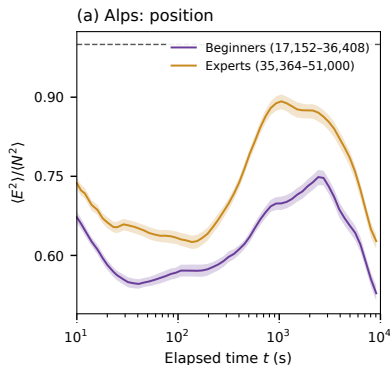
Elapsed time since launch,
not displacement lag.

Position ratio approaches five
in the Pyrenees. Velocity is
also directionally unbalanced.

These uncentred moments
include mean motion.

Paraglider archive; take-off regions. Pointwise 10–90% site–day bootstrap bands; at least 30 flights / 10 groups at each time. Directional balance is weaker than full isotropy.

The experience contrast depends on region and elapsed time



In the Alps the expert proxy is more balanced. In the Pyrenees the curves cross; there is no uniform expert advantage.

Beginners = EN A/B (violet); experts = EN C/D/CCC (ochre). Equipment is an experience proxy, not an individual skill measurement. Same ratio and clock as the previous slide.

PCA describes preferred axes; wind needs independent information

A rotation preserves distance

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

Choose matched environments and an independent wind estimate before defining an “environment-free” model target.

[Regional PCA and its numerical support](#). Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

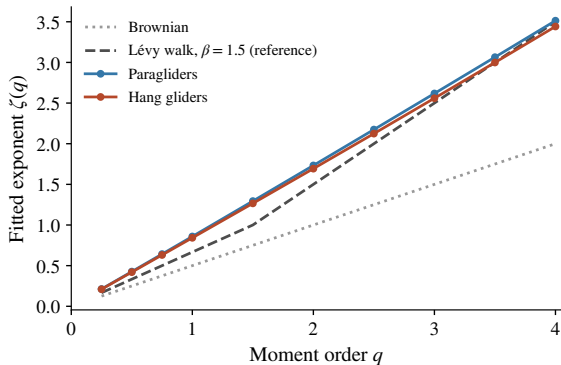
Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

The moment spectrum challenges a specific Lévy-walk benchmark



$$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}.$$

For a constant-speed renewal walk with isotropic independent headings and $\psi(t) \sim t^{-1-\beta}$, $1 < \beta < 2$:

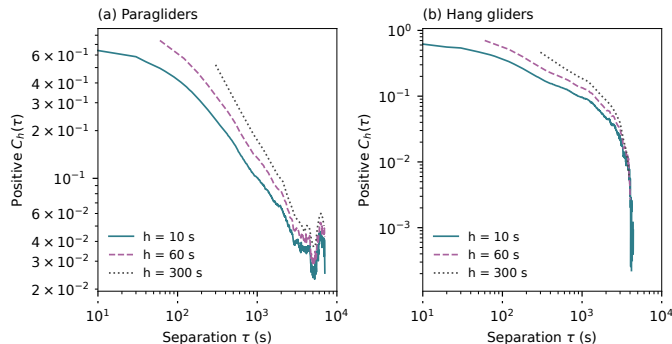
$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

The empirical spectrum is nearly linear over $q = 0.25$ –4.

Can finite-record simulations resolve the predicted branch change under our sampling and selection?

Pooled windows. $\beta = 1.5$ is illustrative, not a fitted null; $q = \beta$ has a logarithmic correction. [Zaburdaev, Denisov & Klafter \(2015\)](#).

Velocity memory persists after averaging over minutes



$$\mathbf{v}_h(t) = \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}.$$

For $h = 10, 60, 300$ s.

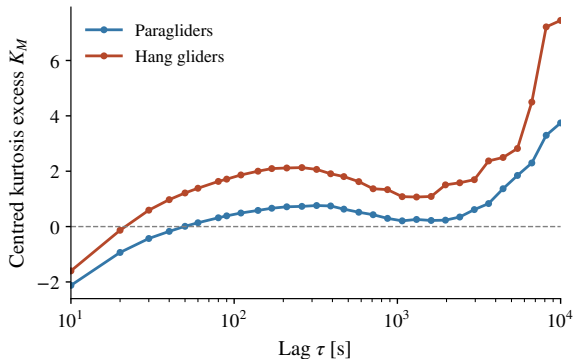
Mean of per-flight centred, normalized correlations.

At 600 s, para correlations are 0.144, 0.198, 0.272 for the three widths.

A finite-persistence model remains plausible: fit its MSD and coarse velocity correlations jointly.

Teal: 10 s; plum dashed: 60 s; grey dotted: 300 s. Positive values shown; [signed curves in backup](#). Curvature prevents a single decay-law claim.

Non-Gaussian pooled increments can arise from Gaussian mixtures



$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \mu),$$

$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

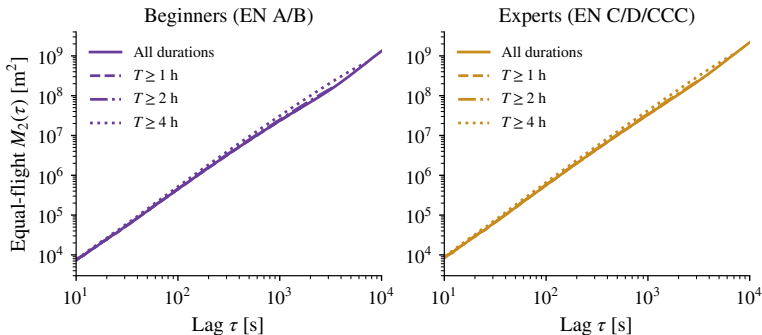
A nonsingular 2D Gaussian has $K_M = 0$, including anisotropic Gaussians.

The observed curve changes sign and rises at large lag, where support is weakest.

Test Gaussianity within matched environments or phases before rejecting conditional Gaussian dynamics.

Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. [Mardia \(1970\)](#).

Long flights occur in both equipment groups

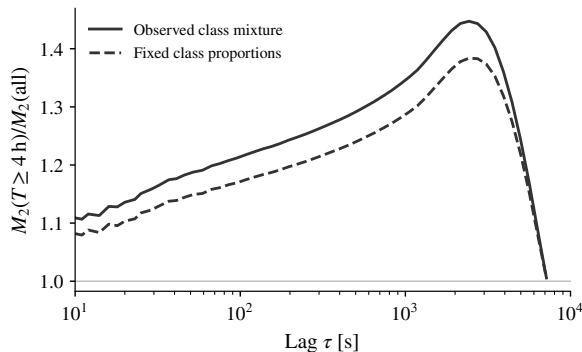


For $T \geq 4$ h: **7,426 beginners** and **27,842 experts**.

Across 10–10,000 s, all-duration slopes are **1.728** and **1.788**.

Equal-flight MSD; groups are nested. Threshold curves stop at $\tau = T_0/2$; at least 30 flights per lag. [All counts and common-range slopes](#).

Fixing equipment proportions removes part of the duration contrast



$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

Fixed weights: $\pi_{A/B} = 0.365$,
 $\pi_{C/D/CCC} = 0.635$.

At $\tau = 1695$ s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

Match site–day and task next. Convergence near 7200 s also reflects shorter flights leaving the reference.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

Which models are challenged, and which remain open?

Candidate	Prediction to test	Present conclusion
Homogeneous Brownian	Linear MSD; independent increments	Challenged over the measured range.
Constant drift + Brownian	Drift-free $V_2 \propto \tau$	Simple version challenged; mixtures unresolved.
Homogeneous fBm	Gaussian law; common H_p	Challenged as one pooled law.
Exponential velocity memory	Ballistic–diffusive crossover	Remains a candidate.
Renewal Lévy walk, $1 < \beta < 2$	Two-branch moment spectrum	Finite-record comparison still needed.

“Challenged” means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

Three decisions for this meeting

1. Define the model target.

Ground-relative motion conditional on environment, or a residual after an independently justified wind correction?

2. Agree the first calibrated comparisons.

Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.

3. Choose the most informative next measurements.

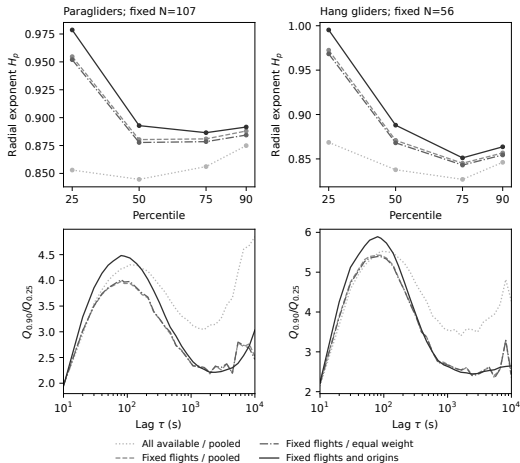
Site-day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

Discussion material

Extra figures, derivations and references

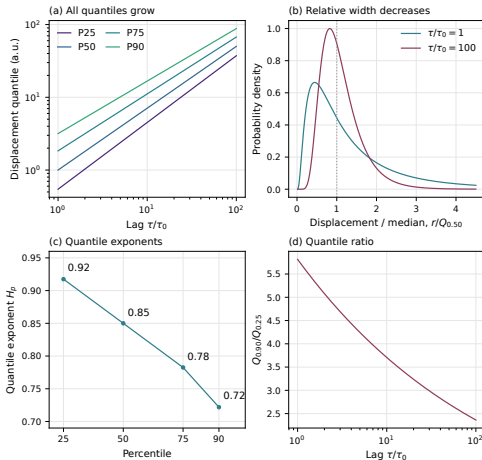
The four population controls



1. All available flights, pooled windows.
2. Fixed long flights, pooled windows.
3. Fixed flights, equal flight weights.
4. Same flights, weights and origins.

$$\hat{F}_\tau(x) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_k \mathbf{1}\{R_{fk}(\tau) \leq x\}.$$

Overlapping origins remain dependent.



$$R(\tau) = (\tau/\tau_0)^{.85} e^{\sigma(\tau)Z}, \quad Z \sim N(0,1),$$

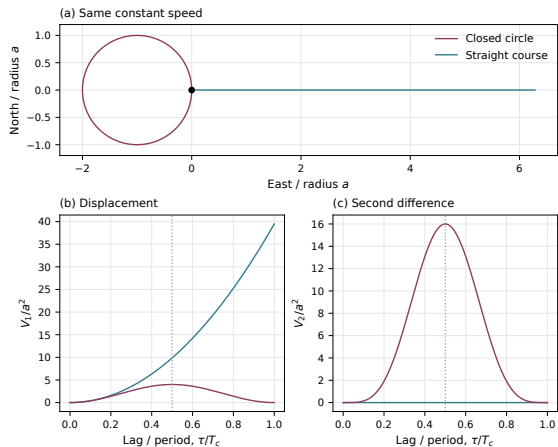
$$\sigma(\tau) = .9 - .1 \log(\tau/\tau_0),$$

$$H_p = .85 - .1 \Phi^{-1}(p).$$

All shown quantiles grow, while spread relative to the median shrinks.

Analytical illustration only; this is not a fitted model for flights.

A circle retains its geometry after second differencing



$$\mathbf{r}(t) = a(\cos \omega t, \sin \omega t), \quad T_c = 2\pi/\omega,$$

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c),$$

$$V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$

The periodic continuation defines V_2 across the illustrated lags. Constant speed does not imply ballistic displacement at every lag.

Exact geometrical example, not an empirical estimate.

Signed memory and the finite-persistence alternative

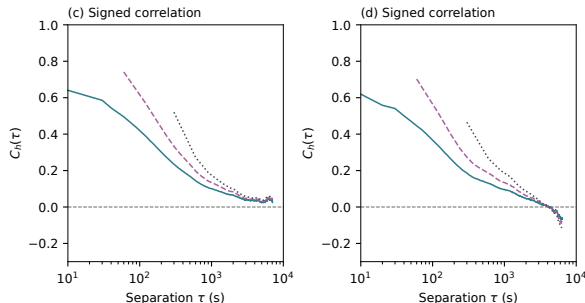
For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds.$$

With $C(s) = e^{-s/t_p}$,

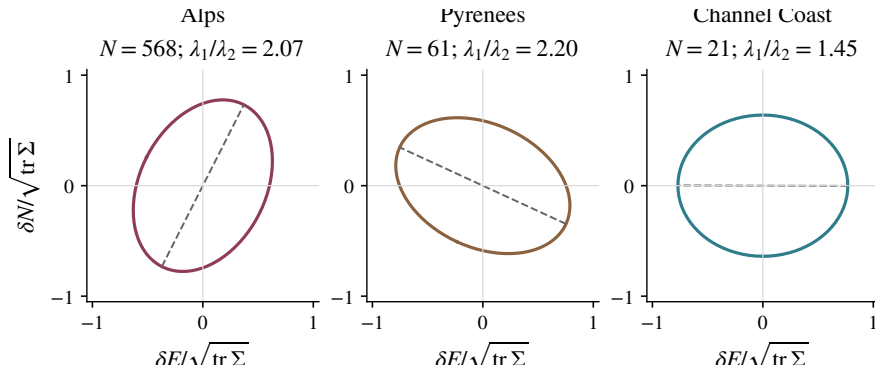
$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.



[Taylor \(1922\)](#), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Regional principal axes are lag-dependent descriptive measurements



Paragliders at $\tau = 1070$ s. Covariance ellipses are normalized by their trace to compare shape and direction, without changing their eigenvalue ratios.

Centred covariance of pooled windows; dashed line = principal axis modulo 180° . Ellipses describe second moments, not Gaussian confidence regions. The coast is a small subgroup and is not an established isotropic control.

Duration and equipment on the common 10–1695 s interval

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,258	1.755	0.022
Beginners	$T \geq 1$ h	51,126	1.754	0.022
Beginners	$T \geq 2$ h	31,779	1.767	0.020
Beginners	$T \geq 4$ h	7,426	1.806	0.017
Experts	All	94,222	1.803	0.017
Experts	$T \geq 1$ h	91,174	1.802	0.017
Experts	$T \geq 2$ h	69,098	1.811	0.016
Experts	$T \geq 4$ h	27,842	1.836	0.015

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.

Sources and the scope of the evidence

Zaburdaev, Denisov & Klafter (2015), *Lévy walks*

Moment-spectrum benchmark

Mandelbrot & Van Ness (1968), *Fractional Brownian motions*

fBm construction

Taylor (1922), *Diffusion by continuous movements*

Velocity/displacement relation

Mardia (1970), *Measures of multivariate skewness and kurtosis*

Centred kurtosis

Efron (1979), *Bootstrap methods*

Resampling principle

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.